MOHAMED MOSTAFA HELAL

AI & Machine Learning Engineer | LLMs, Agentic Systems & Computer Vision

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PROFESSIONAL SUMMARY

AI & Machine Learning Engineer and 2026 Computer Science graduate (GPA 3.62/4.0, Honors) specialized in LLM agentic systems, edge inference, and real-time computer vision. Developed AitroniX 1s at an AI startup under senior engineering direction—an open-source 2.6B bilingual reasoning SLM achieving **68.75% strict AST match on BFCL v4**, optimized for mobile edge deployment via ~1.56 GB GGUF. Proven hands-on track record building local hybrid RAG architectures (FAISS + BM25) and deploying hardware-accelerated multi-stream video analytics pipelines running at **30+ FPS**.

PROFESSIONAL EXPERIENCE

Aitronix (AI Startup) — Remote

AI Engineer — Jan 2026 – Present

- **AitroniX 1s Model Fine-Tuning:** Collaborated under Senior AI Engineer to fine-tune and release AitroniX 1s, an open-source 2.6B bilingual (Arabic/English) reasoning SLM on a hybrid Conv+Attention architecture with 128K context using Unsloth and LoRA/QLoRA.
- **Structured Tool-Calling Evaluation:** Benchmarked structured tool calling on Berkeley Function-Calling Leaderboard (BFCL v4, 400 problems), achieving a **68.75% strict AST match** and 100% valid tool syntax (vs. Qwen3.5-9B at 60.13% and Gemma-4-8B at 48.39%).
- **Edge Quantization & Model Serving:** Quantized model weights into Q4_K_M GGUF (~1.56 GB memory footprint) for private on-device mobile execution and configured high-throughput server deployment using vLLM and SGLang.
- **Real-Time Video Analytics:** Engineered hardware-accelerated computer vision pipelines and multi-camera RTSP streaming engines, sustaining **30+ FPS** across concurrent live feeds via OpenCL GPU acceleration and asynchronous worker queues.

Drug Control & Addiction Prevention Fund — Egypt

Full-Stack Developer — Mutatawi' Rasmi (متطوع رسمي) — Jan 2024 – Dec 2025

- **End-to-End System Development:** Developed the Mutatawi' Rasmi web ecosystem from scratch, implementing responsive user interfaces, backend business logic, and relational database schemas to coordinate nationwide anti-drug youth initiatives.
- **Geospatial Hotspot Analytics:** Engineered data-driven geospatial hotspot analytics and risk visualization maps across Egyptian governorates, empowering administrators to track field missions and allocate operational resources.
- **Volunteer Operations & Engagement:** Implemented volunteer engagement modules, automated leaderboards, milestone verification workflows, and mission tracking modules to manage volunteer participation.

Samsung Innovation Campus — Egypt

AI & Data Science Resident Intern — Jul 2024 – Oct 2024

- **Quantitative Feature Engineering:** Co-developed a quantitative time-series forecasting framework by processing raw market data and engineering 15 financial features, including rolling standard deviations, momentum indices, and historical lags.
- **Predictive Modeling & Optimization:** Constructed and tuned ensemble regression and time-series models (XGBoost, Random Forest, and SARIMA), systematically reducing Mean Absolute Percentage Error (MAPE) from **8.3% to 6.5%**.
- **Exploratory Data Analysis & Dashboards:** Conducted extensive exploratory data analysis (EDA) on customer retention datasets using Pandas and designed interactive dashboards with Plotly and Seaborn to track operational attrition trends.

FEATURED TECHNICAL PROJECTS

AitroniX 1s: Bilingual 2.6B Agentic Reasoning SLM — Hugging Face Model | Jan 2026

Tech: Hybrid Conv+Attention Architecture, GGUF (Q4_K_M), Unsloth, BFCL v4 AST, vLLM, SGLang, llama.cpp

- **Architecture & Open-Source Release:** Engineered an open-source 2.6B bilingual (Arabic/English) reasoning model featuring a hybrid Conv+Attention architecture, 128K context window, and native `<think>` Chain-of-Thought reasoning.
- **Benchmark Evaluation:** Evaluated on Berkeley Function-Calling Leaderboard (BFCL v4 AST, 400 problems), achieving **68.75% strict AST match** with 100% valid tool syntax (vs. Qwen3.5-9B at 60.13%, Gemma-4-8B at 48.39%, and Gemma-4-2B at 35.98%).
- **Edge Inference:** Benchmarked 77.83 on ToolSandbox, 85.49% on IFStruct, 51.87% on AIME25 math, and quantized weights to Q4_K_M GGUF (~1.56 GB) for zero-latency private inference on mobile edge devices.

HyperRAG: Local Hybrid RAG Engine (BM25 + FAISS with Gemma 4 E4B) — Generative AI & Search | 2026

Tech: FAISS, BM25, Reciprocal Rank Fusion (RRF), llama.cpp, Gemma 4 E4B, Python

- **Hybrid Retrieval Pipeline:** Engineered a local hybrid retrieval pipeline combining dense vector search (**FAISS**) with sparse lexical keyword search (**BM25**) via **Reciprocal Rank Fusion (RRF)**, eliminating keyword entity retrieval misses.
- **Private Offline Inference:** Deployed local LLM generation via llama.cpp using Gemma 4 E4B for private, zero-cloud inference, featuring dynamic 800-token chunking (120-token overlap) and verifiable document citation tracking.

Aitronix Real-Time High-FPS Face Recognition Engine — Computer Vision & Systems | 2026

Tech: Python, OpenCV, YuNet, SFace, OpenCL GPU Acceleration, Multithreading, RTSP, LibJPEG

- **High-Speed Face Recognition:** Built a real-time face recognition pipeline utilizing YuNet for single-pass tensor face detection (~2.5ms inference via OpenCL GPU acceleration) and SFace for 128D embeddings with Cosine Similarity vector matching.
- **Multithreading & Byte Decoding:** Designed non-blocking asynchronous multithreading with decoupled worker queues delivering sustained **30+ FPS**; implemented custom native LibJPEG byte decoder to eliminate network buffering delay over RTSP and USB feeds.

FEATURED TECHNICAL PROJECTS (CONTINUED)

CamScence: Multi-Camera Edge Traffic Analytics System — Video Analytics & Edge Vision | 2026

Tech: YOLO (YOLO26n), RTSP Multi-Streaming (11+ Cameras), Speed Estimation, Spatial Heatmaps, Python

- **Multi-Camera Edge Infrastructure:** Architected an edge video analytics system concurrently ingesting **11+ RTSP IP camera streams** with dynamic multi-grid switching and automated feed monitoring.
- **Tracking & Speed Estimation:** Implemented real-time vehicle detection, persistent multi-object tracking, automated vehicle speed estimation (**km/h**), and spatial heatmap overlays for traffic congestion and bottleneck analysis.

Real-Time Video Surveillance Threat & Burglary Detection — Vision-Language Models & Security | 2025 – 2026

Tech: Microsoft Florence-2, OpenAI CLIP, OpenCV, Zero-Shot Grounding, Python

- **Multimodal Video Analysis:** Developed an automated video surveillance pipeline combining Vision-Language Models (**Microsoft Florence-2, OpenAI CLIP**) and OpenCV to detect security incidents, burglary, and physical violence.
- **Zero-Shot Alerting:** Applied zero-shot visual classification and phrase grounding to score threat probabilities frame-by-frame and trigger dynamic real-time on-screen security alerts.

Open-Vocabulary Real-Time Detection & Tracking (YOLO-World) — Computer Vision & Zero-Shot AI | 2025 – 2026

Tech: YOLO-World, Open-Vocabulary Vision, PyTorch, Real-Time GPU Tracking

- **Promptable Zero-Shot Tracking:** Deployed YOLO-World to detect and track custom arbitrary natural language prompted visual concepts without model retraining or additional labeled data.
- **High-FPS GPU Execution:** Optimized local GPU inference pipelines to execute real-time object tracking at **70+ FPS** on high-speed video feeds while maintaining continuous spatial ID tracking.

Integrated AI Ecosystem for Arabian Horses (Graduation Project) — Full-Stack & AI Engineer | Sep 2025 – Jun 2026

Tech: React.js, ASP.NET Core REST API, SQL Server, Python, ConvNeXt, PyTorch

- **Full-Stack Platform:** Architected an end-to-end multi-role web platform (React frontend, ASP.NET Core REST API backend, SQL Server) for horse auctions, farm management (Marabet), and medical record tracking.
- **Deep Learning Pedigree & Valuation:** Integrated a Python deep learning pipeline utilizing **ConvNeXt** for horse pedigree and strain identification alongside regression models for market valuation and price estimation (**Grade: Excellent**).

Additional Projects: NLP Benchmark (DistilBERT 91.36% on 50k IMDb reviews vs RNN/LSTM baselines with FP16); Dynamic WhatsApp AI Recruitment Assistant (n8n + Supabase pgvector + WhatsApp Cloud API); Moondream Edge VLM (webcam VQA); Plant Disease Classifier (38-class CNN, 95.4% accuracy).

TECHNICAL SKILLS

LLMs & Agentic Systems:	**LLM Fine-Tuning (Unsloth, LoRA/QLoRA)**, Structured Tool Calling (JSON/XML, AST), Function Calling, GGUF Quantization, vLLM, SGLang, llama.cpp, Ollama, Hugging Face Transformers, Tokenization, Prompt Engineering
Retrieval & Search (RAG):	**Hybrid RAG**, Dense Vector Search (FAISS), Sparse Lexical Indexing (BM25), Reciprocal Rank Fusion (RRF), Vector Databases (pgvector / Supabase)
Computer Vision & Multimodal:	**YOLO (v8, YOLO-World, YOLO26n)**, Vision-Language Models (Florence-2, CLIP, Moondream), Face Recognition (YuNet, SFace), ConvNeXt, 3D CNNs, MobileNet, Transfer Learning, OpenCV, OpenCL GPU Acceleration, RTSP Multi-Streaming, Spatial Heatmaps
Machine Learning & Analytics:	**PyTorch, TensorFlow, Keras**, Scikit-learn, XGBoost, Random Forest, SARIMA, Time-Series Forecasting, Feature Engineering, Pandas, NumPy, Plotly, Seaborn
Backend & Engineering:	**Python, C#, ASP.NET Core Web API**, React.js, JavaScript, PostgreSQL, Microsoft SQL Server, RESTful APIs, Asynchronous Multithreading, n8n Automation, Docker, Git, GitHub
Languages:	Arabic (Native / Fluent), English (Professional / Fluent)

EDUCATION

Misr University for Science and Technology (MUST) — Giza, Egypt

Bachelor of Science in Computer Science, Major in Artificial Intelligence — Graduated: June 2026

Cumulative GPA: 3.62 / 4.0 (Excellent with Honors) **Graduation Project:** Integrated AI Ecosystem for Arabian Horses (**Grade: Excellent**)

CERTIFICATIONS & PROFESSIONAL CREDENTIALS

Samsung Innovation Campus Certified Professional in Artificial Intelligence & Data Science — Issued: 2024

Issued by Samsung Electronics & Ministry of Communications and Information Technology (MCIT) • Machine Learning, Deep Learning, Statistical Modeling, and Computer Vision

Deep Learning & Computer Vision Specialization — Credential

Comprehensive study in Convolutional Neural Networks (CNNs), Object Detection, and Transfer Learning Architectures

Machine Learning & Advanced Statistical Modeling — Credential

Time-series forecasting, quantitative feature engineering, regression, and tree-based ensemble models